Information-Theoretic Limits and Algorithmic Gaps in Non-Adaptive Group Testing

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Abstract

Group testing seeks to identify a small defective subset within a large population using pooled binary tests. Under the sparse scaling $k = \Theta(n^\theta)$ with $\theta \in (0, 1)$, non-adaptive noiseless group testing exhibits sharp threshold phenomena: exact recovery becomes possible once the number of tests exceeds a critical value determined by θ.

Recent results have characterized these information-theoretic thresholds for classical random designs, including Bernoulli and near-constant column weight constructions. However, efficient decoding algorithms may require strictly more tests than the information-theoretic minimum, giving rise to an *algorithmic gap* between statistical feasibility and computational tractability.

This paper presents a unified exposition of information-theoretic and algorithmic thresholds in the sparse non-adaptive setting. We compare practical decoders across test designs, formalize the algorithmic gap in terms of normalized test complexity, and explain its structural origin in the geometry of the underlying bipartite test graph. We then highlight recent structured constructions that establish sharp information-theoretic and algorithmic phase transitions, demonstrating that the apparent computational hardness of non-adaptive noiseless group testing is design-dependent rather than intrinsic.

While spatial structure removes the computational barrier in the noiseless setting, an adaptivity gap persists in certain sparsity regimes. Overall, the modern theory reveals a delicate interplay between sparsity, design geometry, and algorithmic performance in group testing.

2012 ACM Subject Classification Theory of computation → Design and analysis of algorithms; Mathematics of computing → Information theory

Keywords and phrases Group Testing, Non-Adaptive Algorithms, Information-Theoretic Limits, Algorithmic Gap, Spatial Coupling

Digital Object Identifier 10.4230/LIPIcs...

1 Introduction

Group testing studies the problem of identifying a small subset of defective items within a large population using pooled tests. A test is positive if it contains at least one defective item and negative otherwise. Originally introduced by Dorfman in 1943 for blood screening [9], group testing has since found applications in molecular biology and DNA sequencing, medical diagnostics and veterinary surveillance [13], large-scale pooled COVID-19 testing [10, 1], as well as network tomography and large-scale data processing [3].

In many modern applications, tests must be designed in parallel. This leads to the *non-adaptive* group testing model, where the test matrix is fixed in advance. While adaptive procedures can achieve optimal information efficiency, non-adaptive designs are operationally simpler and better suited to large-scale or time-sensitive settings.

Under the sparse scaling $k = \Theta(n^\theta)$ with $\theta \in (0, 1)$, non-adaptive group testing exhibits sharp threshold phenomena. Information-theoretic analyses determine the minimal number of tests required for exact recovery, irrespective of computational complexity. These thresholds have been characterized for Bernoulli test designs and near-constant column weight constructions [2, 6, 3].

However, achieving these limits algorithmically is more delicate. Efficient decoders such as COMP and DD often require strictly more tests than the information-theoretic minimum. This separation between statistical feasibility and computational feasibility is commonly referred to as the *algorithmic gap* [3]. Recent works have shown that this gap depends strongly on the underlying test design and its induced combinatorial geometry.

The goal of this paper is to provide a structured and unified exposition of recent advances in non-adaptive noiseless group testing. We focus on three intertwined aspects:

- information-theoretic thresholds,
- algorithmic thresholds of practical decoders,
- the structural origin and resolution of the algorithmic gap.

We present the sharp capacity results for Bernoulli and near-constant column weight designs in a common framework, expressing thresholds explicitly in terms of the sparsity parameter θ. We then compare these limits with the performance of efficient decoding algorithms, highlighting the regimes in which polynomial-time recovery matches or fails to match information-theoretic optimality.

Finally, we discuss recent structured design constructions that establish a sharp information-theoretic and algorithmic phase transitions in the noiseless setting [7], illustrating that computational hardness in non-adaptive group testing is design-dependent rather than intrinsic.

Our aim is not to introduce new threshold results, but to clarify the interplay between information limits, algorithmic performance, and design structure, and to provide a coherent perspective on the modern theory of sparse non-adaptive group testing.

2 Problem Setup and Preliminaries

2.1 Non-Adaptive Group Testing Model

We consider the classical noiseless non-adaptive group testing problem. Let $n \in \mathbb{N}$ denote the number of items and let $S \subseteq [n] := \{1, \ldots, n\}$ be an unknown defective set with $|S| = k$.

In the non-adaptive setting, all tests are specified in advance by a binary matrix

$$G \in \{0,1\}^{m \times n},$$

where m denotes the number of tests. The entry $G_{ij} = 1$ indicates that item j participates in test i. Equivalently, the test design can be viewed as a bipartite graph between the set of items and the set of tests, with an edge connecting item j to test i whenever $G_{ij} = 1$. All tests are performed in parallel and do not depend on previous outcomes.

Under the noiseless OR model, the outcome vector $\hat{\sigma} \in \{0,1\}^m$ is given by

$$\hat{\sigma}_i = \bigvee_{j \in S} G_{ij}, \qquad i = 1, \ldots, m,$$

that is, a test is positive if and only if it contains at least one defective item.

Given $(G, \hat{\sigma})$, the goal is to reconstruct S. We say that recovery is successful if an estimator $\hat{S}$ satisfies

$$\mathbb{P}(\hat{S} \neq S) \to 0 \quad \text{as } n \to \infty.$$

Throughout the paper, we study the asymptotic regime $n \to \infty$. We focus exclusively on the non-adaptive setting; adaptive strategies, where tests may depend sequentially on previous outcomes (e.g., [9, 11, 4]), are not considered here.

2.2 Sparse Regime

We consider the standard sparse scaling in which the number of defectives grows sublinearly with the population size. Specifically, we assume

$$k = \Theta(n^\theta), \qquad \theta \in (0,1),$$

where the parameter θ characterizes the sparsity level.

Throughout the paper, asymptotic statements are understood in the regime $n \to \infty$ under this scaling. All thresholds and rate expressions will be stated as functions of θ.

2.3 Test Design Ensembles

We consider non-adaptive random constructions of the test matrix $G \in \{0,1\}^{m\times n}$. In this work, we focus on two standard random design ensembles. For a broader overview of pooling designs, see [5]. Basic probabilistic properties used in later analyses are summarized in Appendix A.

2.3.1 Bernoulli Design

In the Bernoulli design, the entries of G are drawn independently as

$$G_{ij} \sim \mathrm{Bernoulli}(p),$$

with scaling

$$p = \frac{\nu}{k},$$

where $\nu > 0$ is a design parameter. Equivalently, each item is included in each test independently with probability ν/k (see, e.g., [3]).

2.3.2 Near-Constant Column Weight Design

In the near-constant column weight design, each item is assigned to

$$L = \frac{\nu m}{k}$$

tests chosen independently and uniformly at random with replacement, where $\nu > 0$ is a design parameter. Thus, each column has weight concentrated around L (see, e.g., [6, 3]).

3 Information-Theoretic Limits

3.1 Error Probability

Let $S \subseteq [n]$ denote the defective set with $|S| = k$, and let $\hat{S}$ be the output of a decoder based on the test matrix G and the outcome vector $\hat{\sigma}$. We define the average error probability as

$$P_e = \Pr(\hat{S} \neq S),$$

where the probability is taken over the randomness of the defective set, assumed uniformly distributed over all subsets of size k, as well as any randomness in the test design and decoding procedure.

Exact recovery is said to be achievable if

$$P_e \to 0 \quad \text{as } n \to \infty$$

under the sparse scaling $k = \Theta(n^\theta)$.

3.2 Counting Bound

Under the noiseless non-adaptive model, the decoder must recover the defective set $S \subseteq [n]$ with $|S| = k$ from the outcome vector $Y \in \{0,1\}^m$.

There are

$$\binom{n}{k}$$

possible defective sets. Since each test produces a single binary outcome, m tests can generate at most 2^m distinct outcome vectors. Hence, reliable recovery with vanishing error probability requires

$$2^m \geq \binom{n}{k},$$

which yields the classical counting bound

$$m \geq \log_2 \binom{n}{k}$$

(see, e.g., [3]).

An equivalent converse can be obtained via Fano's inequality [8] by treating S as a random variable. A detailed derivation is provided in Appendix B.

Under the sparse scaling $k = \Theta(n^\theta)$ with $\theta \in (0,1)$, Stirling's approximation gives

$$\log_2 \binom{n}{k} = (1 - \theta + o(1))\, k \log_2 n.$$

3.3 Entropy Interpretation

The counting bound admits a natural information-theoretic interpretation. If the defective set S is drawn uniformly at random over all $\binom{n}{k}$ subsets, then

$$H(S) = \log_2 \binom{n}{k}.$$

Thus, identifying S requires recovering a message of entropy $H(S)$.

In the noiseless setting, each test produces at most one bit of information. Consequently, the total number of tests must scale at least linearly with $H(S)$. This perspective connects non-adaptive group testing to classical channel coding, where a message must be reliably recovered from binary observations [8, 3].

3.4 Sparse Scaling

Under the standard sparse regime

$$k = \Theta(n^\theta), \qquad \theta \in (0,1),$$

Stirling's approximation gives

$$\log_2 \binom{n}{k} = (1 - \theta + o(1))\, k \log_2 n.$$

Combining this with the counting bound yields the fundamental information-theoretic requirement

$$m \geq (1-\theta)\, k \log_2 n\, (1 + o(1)).$$

Thus, in the noiseless non-adaptive setting, the number of tests must scale as

$$m = \Omega(k \log n),$$

with leading constant $(1-\theta)$.

3.5 Rate

The rate of a group testing scheme is defined as

$$R = \frac{H(S)}{m} = \frac{\log_2 \binom{n}{k}}{m},$$

which measures the number of information bits resolved per test.

From the counting bound, any scheme achieving vanishing error probability in the noiseless setting must satisfy

$$R \leq 1.$$

Under the sparse scaling $k = \Theta(n^{\theta})$, this corresponds to

$$R = \frac{(1-\theta) k \log_2 n}{m} + o(1).$$

3.6 Converse and Achievability

A converse result shows that exact recovery is impossible when m falls below the counting bound, while an achievability result establishes that this threshold can be attained by some (not necessarily computationally efficient) decoding rule (see, e.g., [8, 3]). When these bounds coincide up to lower-order terms, the information-theoretic threshold is determined. Importantly, such achievability results do not imply that polynomial-time algorithms achieve the same performance.

4 Information-Theoretic Thresholds

4.1 Bernoulli Design

We characterize the sharp information-theoretic threshold for Bernoulli non-adaptive testing under the sparse regime introduced earlier. All statements are asymptotic as $n \to \infty$, and exact recovery is required with probability tending to one.

4.1.1 Capacity

The information-theoretic capacity of Bernoulli non-adaptive group testing in the sparse regime is given by [2]

$$C(\theta) = \max_{\nu > 0} \min \left\{ \frac{\nu e^{-\nu}}{\ln 2} \frac{1-\theta}{\theta},\ h(e^{-\nu}) \right\},$$

where $h(\cdot)$ denotes the binary entropy function.

The two terms inside the minimum correspond to distinct information bottlenecks:

- The term

  $$\frac{\nu e^{-\nu}}{\ln 2}\frac{1-\theta}{\theta}$$

  reflects an *isolation constraint*: each defective must appear alone in sufficiently many tests to avoid structural ambiguity.
- The term

  $$h(e^{-\nu})$$

  reflects an *entropy constraint*: the global distribution of test outcomes must carry sufficient information to identify the defective set.

The capacity arises by balancing these two effects.

4.1.2 Converse and Achievability

Scarlett and Cevher [15] proved that the rate $C(\theta)$ is achievable. Equivalently, exact recovery is possible if

$$m > (1+\varepsilon)\frac{\log_2\binom{n}{k}}{C(\theta)}$$

for any fixed $\varepsilon > 0$.

A matching converse was established in [2], showing that if

$$m < (1-\varepsilon)\frac{\log_2\binom{n}{k}}{C(\theta)},$$

then the error probability is bounded away from zero. Hence, the threshold is sharp up to lower-order terms.

4.1.3 Threshold in Terms of Number of Tests

Using the asymptotic expansion

$$\log_2\binom{n}{k} = (1-\theta+o(1))\,k\log_2 n,$$

the recovery threshold can be written as

$$m^*(\theta) = (1+o(1))\,c_{\mathrm{Bern}}(\theta)\,k\log_2 n,$$

where

$$c_{\mathrm{Bern}}(\theta) = \frac{1-\theta}{C(\theta)}.$$

4.1.4 Regime Structure

The behavior of $C(\theta)$ exhibits distinct regimes.

Regime I: $\theta \le 1/3$.

The maximization is achieved at $\nu = \ln 2$, the entropy constraint dominates, and

$$C(\theta) = 1.$$

Hence,

$$m^*(\theta) = (1-\theta)\, k \log_2 n,$$

which matches the counting bound. In this regime, Bernoulli non-adaptive testing is *information-theoretically optimal.*

Regime II: $\theta > 1/3$.

For larger θ, the isolation constraint becomes active and $C(\theta) < 1$, so Bernoulli testing becomes *strictly suboptimal in the information-theoretic sense.* The fundamental phase transition therefore occurs at $\theta = 1/3$, where the capacity drops below 1.

In the interval $1/3 < \theta < \theta^* \approx 0.359$ (see [2]), the optimizer ν is determined by balancing the two terms in the max–min expression. For $\theta \ge \theta^*$, the optimizer satisfies $\nu = 1$, and the capacity simplifies to

$$C(\theta) = \frac{1}{e \ln 2} \frac{1-\theta}{\theta}.$$

Consequently,

$$m^*(\theta) = (1+o(1))\, e \ln 2\, \theta\, k \log_2 n.$$

4.2 Near-Constant Column Weight Design

We now state the sharp information-theoretic threshold for the near-constant column weight (NCC) design under the sparse regime introduced earlier.

4.2.1 Capacity

The information-theoretic threshold for the NCC design was established in [6, Theorem I.1]. The number of tests required for exact recovery satisfies

$$m_{\inf}(n,\theta) = k \log_2(n/k) \min\left\{1, \frac{1-\theta}{\theta \log 2}\right\}.$$

Using the asymptotic identity

$$\log_2(n/k) = (1-\theta) \log_2 n,$$

this can be written as

$$m_{\inf}(n,\theta) = (1-\theta)\, k \log_2 n \min\left\{1, \frac{1-\theta}{\theta \log 2}\right\}.$$

Equivalently, the capacity of the NCC design is

$$C_{\mathrm{NCC}}(\theta) = \max\left\{1, \frac{\theta \log 2}{1-\theta}\right\}.$$

4.2.2 Converse and Achievability

The threshold is sharp in the following sense [6, Theorem I.1]:

- If
  $$m < (1-\varepsilon)m_{\text{inf}}(n,\theta),$$
  then exact recovery is impossible with vanishing error probability.
- If
  $$m > (1+\varepsilon)m_{\text{inf}}(n,\theta),$$
  then there exists a (possibly computationally unbounded) decoder that succeeds with high probability.

Thus, the information-theoretic threshold is determined up to lower-order terms.

4.2.3 Threshold in Terms of Number of Tests

Using

$$\log_2 \binom{n}{k} = (1-\theta+o(1))\, k \log_2 n,$$

the recovery threshold can be written as

$$m^*_{\text{NCC}}(\theta) = (1+o(1))\, c_{\text{NCC}}(\theta)\, k \log_2 n,$$

where

$$c_{\text{NCC}}(\theta) = \frac{1-\theta}{C_{\text{NCC}}(\theta)}.$$

4.2.4 Regime Structure

The behavior of $C_{\text{NCC}}(\theta)$ exhibits two regimes.

Regime I: $\theta \le \frac{\log 2}{1+\log 2} \approx 0.41$.

In this regime,

$$C_{\text{NCC}}(\theta) = 1,$$

and hence

$$m^*_{\text{NCC}}(\theta) = (1-\theta)\, k \log_2 n,$$

which matches the counting bound. Thus, the near-constant design is information-theoretically optimal in this sparsity regime.

Regime II: $\theta > \frac{\log 2}{1+\log 2}$.

In this regime,

$$C_{\mathrm{NCC}}(\theta) = \frac{\theta \log 2}{1-\theta},$$

and therefore

$$m^*_{\mathrm{NCC}}(\theta) = (1+o(1))\,\frac{(1-\theta)^2}{\theta \log 2}\, k \log_2 n.$$

Hence, for larger values of θ, the near-constant design becomes strictly suboptimal in the information-theoretic sense, as the achievable rate falls below one.

The transition at

$$\theta = \frac{\log 2}{1+\log 2}$$

marks the point at which the second term in the capacity expression becomes active.

4.3 Comparison of Bernoulli and Near-Constant Designs

We now compare the information-theoretic thresholds of the Bernoulli and near-constant column weight (NCC) designs under the sparse scaling $k = \Theta(n^\theta)$. Figure 1 illustrates the normalized test complexity

$$c(\theta) = \frac{m}{k \log_2 n}$$

required for exact recovery under both designs, together with the universal counting bound.

A first difference appears in the range of parameters for which each design attains the counting bound. For the Bernoulli design, the achievable rate equals the counting bound (i.e., $C(\theta) = 1$) for $\theta \le 1/3$, and drops strictly below 1 once $\theta > 1/3$. Thus, Bernoulli testing is information-theoretically optimal only in the regime $\theta \le 1/3$. In contrast, the NCC design achieves its maximal non-adaptive rate up to

$$\theta \le \frac{\log 2}{1+\log 2} \approx 0.41,$$

and becomes suboptimal only beyond this point. Hence, the NCC design remains information-theoretically optimal in a strictly larger sparsity regime.

The earlier breakdown of optimality for Bernoulli testing can be traced to fluctuations in column weights. As θ increases, the probability that a defective item is insufficiently isolated across tests grows, and the isolation constraint becomes dominant (see, e.g., [3]). The NCC design mitigates this effect by enforcing near-regular column degrees, which reduces variability in item participation and delays the onset of the isolation bottleneck. Consequently, the associated phase transition occurs at a larger value of θ, as illustrated in Figure 1.

For larger values of θ, both designs become strictly suboptimal in the information-theoretic sense, and the required number of tests scales as

$$m^*(\theta) = (1+o(1))\, c(\theta)\, k \log_2 n,$$

with different θ-dependent constants. In this regime, recovery is governed primarily by structural distinguishability constraints rather than entropy considerations alone.

Overall, the NCC design strictly improves the information-theoretic threshold over Bernoulli testing across an intermediate range of sparsity parameters. This structural advantage will play a central role in understanding the gap between information-theoretic and algorithmic limits discussed in the next section.

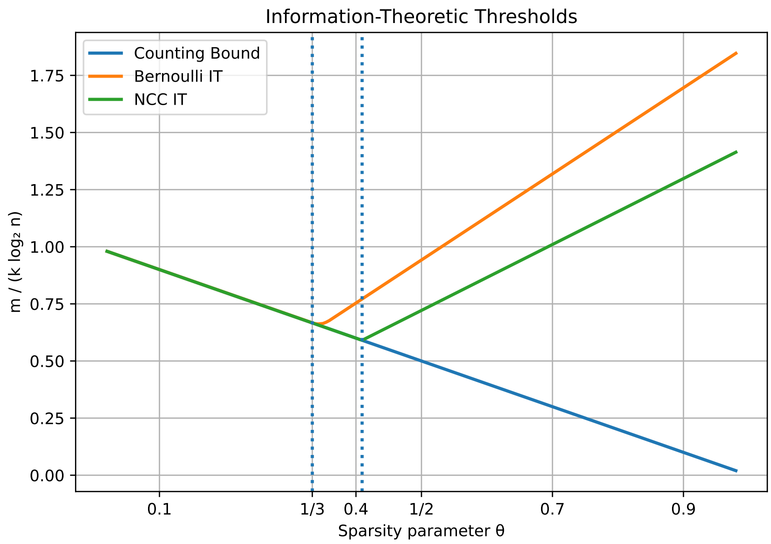

Figure 1 Information-theoretic thresholds under Bernoulli and near-constant column weight (NCC) designs. The vertical axis shows the normalized test complexity $c(\theta) = m/(k \log_2 n)$. The dotted vertical lines indicate the respective phase transitions: $\theta = 1/3$ for Bernoulli testing and $\theta = \log 2/(1 + \log 2)$ for NCC.

5 Algorithmic Thresholds

We now investigate the performance of polynomial-time decoding algorithms under the two random design ensembles considered above. Throughout, we focus on exact recovery under the sparse scaling $k = \Theta(n^\theta)$. The decoding algorithms considered here, along with their asymptotic performance guarantees, are classical in the group testing literature; see, e.g., [3] for a comprehensive treatment.

5.1 Algorithmic Recovery Criterion

For a given decoding algorithm $\mathcal{A}$, we say that it achieves exact recovery if its error probability tends to zero as $n \to \infty$. An *algorithmic threshold* is a constant $c_{\mathcal{A}}(\theta)$ such that

$$m \geq (1 + o(1))\, c_{\mathcal{A}}(\theta)\, k \log_2 n$$

is sufficient for successful recovery by $\mathcal{A}$, while recovery fails below this scaling.

We consider the following algorithms:

- *COMP:* declares any item appearing in a negative test as non-defective and all remaining items as defective.
- *DD (Definite Defectives):* runs COMP to eliminate definite non-defectives, then declares items that appear alone in a positive test as defective.
- *SCOMP:* extends DD by greedily selecting items to cover unexplained positive tests.
- *SSS (Smallest Satisfying Set):* outputs the smallest set consistent with all outcomes; statistically optimal but computationally intractable.

COMP, DD, and SCOMP run in polynomial time, while SSS is exponential-time and serves as a statistical benchmark.

5.2 Thresholds under the Bernoulli Design

We first consider the Bernoulli design with inclusion probability $p = \nu/k$.

5.2.1 COMP

Under Bernoulli testing, COMP succeeds provided

$$m \geq (1 + o(1)) \frac{e \ln 2}{1 - \theta} k \log_2 n.$$

COMP never achieves the Bernoulli achievable rate and is strictly suboptimal for all θ.

5.2.2 DD

The DD algorithm improves upon COMP by exploiting singleton-positive tests. Its threshold satisfies

$$m \geq (1 + o(1)) \, c_{\mathrm{DD,Bern}}(\theta) \, k \log_2 n,$$

where $c_{\mathrm{DD,Bern}}(\theta)$ coincides with the Bernoulli achievable-rate constant

$$\frac{1 - \theta}{R_{\mathrm{Bern}}(\theta)}$$

for $\theta \geq 1/2$, and is strictly larger for smaller θ.

Thus DD is information-theoretically optimal within the Bernoulli design for $\theta \geq 1/2$, and suboptimal otherwise.

5.2.3 SCOMP and SSS

SCOMP matches the DD threshold asymptotically and does not improve the scaling constant.

The SSS decoder achieves the Bernoulli achievable rate

$$R_{\mathrm{Bern}}(\theta) = \max_{\nu > 0} \min \left\{ h(e^{-\nu}), \frac{\nu e^{-\nu}}{\ln 2} \frac{1 - \theta}{\theta} \right\},$$

and therefore matches the information-theoretic limit for the Bernoulli design. However, it requires exponential time.

5.3 Thresholds under the Near-Constant Column Weight Design

We now consider the near-constant column weight (NCC) design.

5.3.1 COMP

Under the NCC design, COMP succeeds if

$$m \geq (1 + o(1)) \, c_{\mathrm{COMP,NCC}}(\theta) \, k \log_2 n,$$

where $c_{\mathrm{COMP,NCC}}(\theta) = 1/\ln 2$. This represents a strict improvement over the Bernoulli constant, but COMP remains information-theoretically suboptimal for all θ.

5.3.2 DD

The DD algorithm achieves

$$m \geq (1+o(1))\, c_{\mathrm{DD,NCC}}(\theta)\, k \log_2 n.$$

Under the NCC design, DD matches the achievable rate

$$R_{\mathrm{NCC}}(\theta) = \min\left\{1, (\ln 2)\frac{1-\theta}{\theta}\right\}$$

for $\theta \geq 1/2$. Thus, the structural regularity of the NCC design enlarges the regime of algorithmic optimality.

5.3.3 SCOMP and SSS

As in the Bernoulli case, SCOMP matches DD asymptotically.

The SSS decoder achieves the full NCC achievable rate for all θ, but remains computationally infeasible.

5.4 Comparison with Information-Theoretic Limits

The preceding results reveal that statistical optimality and computational efficiency need not coincide. We now make this separation explicit by comparing the algorithmic thresholds of efficient decoders with the corresponding information-theoretic limits. In particular, we distinguish between the universal counting bound, which applies to all designs, and the sharper design-dependent achievable rates characterized earlier.

Under the Bernoulli design, the DD algorithm matches the design-dependent information-theoretic threshold only for $\theta \geq 1/2$, while for $\theta < 1/2$ a strict separation between information-theoretic and polynomial-time recovery persists.

Under the near-constant column weight design, the achievable constants improve, and the regime of algorithmic optimality expands, though a gap remains in part of the sparsity range.

The SSS decoder matches the information-theoretic threshold for each design, but requires exponential time.

5.5 The Algorithmic Gap

We formalize the separation between statistical optimality and computational feasibility in terms of the normalized test complexity

$$c(\theta) = \frac{m}{k \log_2 n}.$$

Information-theoretic benchmark.

The fundamental limit for exact recovery is given by the counting bound

$$m \geq (1-\theta)\, k \log_2 n,$$

which holds independently of the test design. We denote the corresponding constant by

$$c_{\mathrm{count}}(\theta) = 1-\theta.$$

Under the Bernoulli design, the sharper achievable rate is

$$R_{\mathrm{Bern}}(\theta) = \max_{\nu>0} \min \left\{ h(e^{-\nu}), \frac{\nu e^{-\nu}}{\ln 2} \frac{1-\theta}{\theta} \right\},$$

leading to the design-dependent threshold

$$m \geq (1+o(1)) \frac{1-\theta}{R_{\mathrm{Bern}}(\theta)} \, k \log_2 n.$$

Best known efficient algorithm.

For Bernoulli testing, the Definite Defectives (DD) algorithm requires

$$m \geq (1+o(1))\, c_{\mathrm{DD,Bern}}(\theta)\, k \log_2 n,$$

where $c_{\mathrm{DD,Bern}}(\theta) > c_{\mathrm{count}}(\theta)$ for $\theta < 1/2$ [3].

Hence, for $\theta < 1/2$ there exists a nontrivial interval

$$c_{\mathrm{count}}(\theta)\, k \log_2 n < m < c_{\mathrm{DD,Bern}}(\theta)\, k \log_2 n$$

in which exact recovery is information-theoretically possible, but no polynomial-time algorithm is known. Equivalently, in terms of the normalized constant $c(\theta)$, there exists a strict separation between $c_{\mathrm{count}}(\theta)$ and $c_{\mathrm{DD,Bern}}(\theta)$.

This interval constitutes the *algorithmic gap*. The gap is illustrated in Figure 2, where the vertical axis represents the normalized number of tests $c(\theta)$ and the shaded region highlights the range $\theta < 1/2$ in which polynomial-time recovery is strictly suboptimal.

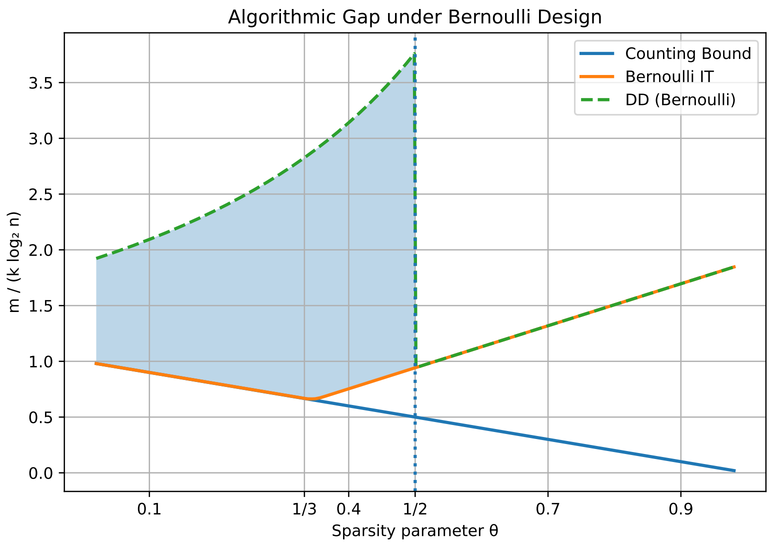

Figure 2 Algorithmic gap under the Bernoulli design. The vertical axis shows the normalized test complexity $c(\theta) = m/(k \log_2 n)$. The shaded region indicates the regime $\theta < 1/2$ in which exact recovery is information-theoretically possible but no polynomial-time algorithm is known. The dotted vertical line marks the transition at $\theta = 1/2$.

5.5.1 Structural Origin

The gap arises from combinatorial obstructions in the random test design.

Under the Bernoulli ensemble, two problematic sets may persist:

- V_0^+: non-defective items appearing only in positive tests,
- V_1^+: defective items never appearing in isolation.

Above the information-theoretic threshold, $V_1^+ = \varnothing$ with high probability. However, V_0^+ may remain large, producing many nearly indistinguishable satisfying configurations.

Recovery can be formulated as selecting a minimum-weight configuration consistent with all test outcomes, which corresponds to a minimum hypergraph vertex cover problem. Such combinatorial optimization problems are NP-hard in general, providing heuristic evidence that the observed gap may reflect an inherent computational barrier.

Effect of Structured Designs.

Under the near-constant column weight design, degree regularity reduces masking effects and shrinks the gap region, though it does not eliminate the computational barrier entirely [6].

6 Resolution of the Algorithmic Gap via Structured Designs

The algorithmic gap identified in Section 5.5 was shown to depend strongly on the geometry of the underlying random test design. Recent work demonstrates that this gap can disappear under carefully structured non-adaptive constructions.

In particular, spatially coupled test designs introduced for group testing in [7], inspired by threshold saturation phenomena in coding theory [12, 14], modify the global geometry of the bipartite test graph and enable progressive recovery.

6.1 Spatial Coupling Principle

The construction introduced in [7] implements spatial coupling in the non-adaptive noiseless setting. Instead of a single global random bipartite graph, the items are partitioned into compartments arranged in a chain or ring. Each item participates only in tests from a bounded number of neighboring compartments. Within each compartment, the design enforces approximate degree regularity, so that each item participates in a nearly constant number of tests within a bounded local window.

This structured construction breaks global symmetry and induces a propagation effect: once recovery succeeds in one region, it provides reliable boundary information for adjacent regions.

6.2 Algorithmic Implications

Under spatially coupled designs, a polynomial-time decoding procedure achieves exact recovery whenever the number of tests exceeds the sharp information-theoretic threshold [7]. More precisely, for noiseless non-adaptive testing under sparse scaling $k = \Theta(n^\theta)$, the algorithmic threshold matches the information-theoretic threshold. Thus, the algorithmic gap observed under classical Bernoulli designs disappears.

Importantly, the computational barrier in the classical setting was not due to lack of information, but to the presence of exponentially many near-satisfying configurations in the solution space. Spatial coupling alters this geometry, removing the hard phase and enforcing progressive recovery.

6.3 Complexity and Broader Implications

The decoding procedures operate in time polynomial in n and m, and avoid solving NP-hard subproblems explicitly. The results are asymptotic in n and currently established for the noiseless setting.

Although the computational gap disappears under spatial coupling, an adaptivity gap remains. For certain sparsity regimes, adaptive strategies require strictly fewer tests than any non-adaptive scheme [3]. Thus, spatial coupling resolves a computational limitation, but does not eliminate the fundamental advantage of adaptivity.

More broadly, these developments indicate that computational hardness in non-adaptive noiseless group testing is design-dependent rather than intrinsic. Structured constructions reshape the geometry of the test matrix, eliminating the metastable phase that obstructs greedy and local algorithms. This clarifies the relationship between information-theoretic limits, algorithmic feasibility, and combinatorial structure in modern group testing theory.

7 Conclusion

We have presented a unified perspective on recent advances in non-adaptive noiseless group testing under sparse scaling. By organizing information-theoretic thresholds, algorithmic performance guarantees, and structural design principles within a common framework, we clarified the interplay between statistical limits and computational feasibility.

Information-theoretic analyses determine the minimal number of tests required for exact recovery, while practical decoding algorithms may require strictly more tests depending on the test design. This separation gives rise to the algorithmic gap. Recent structured constructions demonstrate that this gap is not inherent to non-adaptive group testing, but rather a consequence of specific random ensembles. When the geometry of the test matrix is appropriately modified, polynomial-time recovery can match the information-theoretic threshold in the noiseless non-adaptive setting.

At the same time, an adaptivity gap persists in certain sparsity regimes, highlighting a fundamental distinction between computational barriers and adaptive advantages.

Overall, these developments reveal that the apparent computational hardness of non-adaptive noiseless group testing is intimately tied to the geometry of the test design.

8 Open Problems

Despite the sharp asymptotic characterizations achieved in the noiseless setting, several fundamental questions remain open.

- *Noisy extensions.* While the noiseless information-theoretic and algorithmic thresholds are now well understood, the interaction between structured designs and noise remains less explored. Can spatially structured constructions close the algorithmic gap under standard symmetric noise models? Do analogous threshold saturation phenomena occur in noisy group testing?
- *Finite-length performance.* Most threshold results are asymptotic. Understanding the finite-length behavior of structured designs, including precise constants and practical parameter choices, is important for real-world applications.
- *Robustness to model mismatch.* Structured constructions often rely on carefully tuned parameters. How sensitive are these designs to deviations in sparsity level or test density? Can universal constructions achieve near-optimal performance without precise parameter knowledge?

- *Computational lower bounds.* Under classical random designs, the algorithmic gap suggests an intrinsic computational barrier. Is it possible to formalize this barrier through average-case complexity reductions or conditional hardness assumptions, thereby establishing provable computational thresholds?
- *Beyond exact recovery.* Many applications permit partial recovery or tolerate a small number of errors. How do algorithmic and information-theoretic thresholds interact in approximate recovery regimes?

Addressing these questions would further clarify the relationship between information limits, design structure, and computational complexity in modern group testing theory.

A Probabilistic Properties of the Test Designs

This appendix collects basic probabilistic properties of the random design ensembles used throughout the paper. All asymptotic statements are understood in the regime $n \to \infty$ under the sparse scaling $k = \Theta(n^\theta)$.

A.1 Bernoulli Design

Under the Bernoulli design with inclusion probability $p = \nu/k$, the entries of the test matrix are independent.

Row Weights.

For a fixed test i, the number of participating items follows

$$\mathrm{Binomial}(n, p).$$

If k items are defective, the number of defectives in a fixed test follows

$$\mathrm{Binomial}(k, p),$$

and hence

$$\mathbb{E}[\text{defectives in a test}] = kp = \nu.$$

Negative Test Probability.

The probability that a given test is negative equals

$$\mathbb{P}(\text{test is negative}) = (1-p)^k = \left(1 - \frac{\nu}{k}\right)^k = e^{-\nu} + o(1).$$

Concentration.

Standard Chernoff bounds imply that the number of negative tests concentrates sharply around its expectation

$$me^{-\nu}.$$

These concentration properties are used in the proofs of the information-theoretic and algorithmic thresholds.

A.2 Near-Constant Column Weight Design

In the near-constant column weight design, each item participates in

$$L = \frac{\nu m}{k}$$

tests chosen independently and uniformly at random with replacement.

Column Weights.

Each column is assigned exactly L test selections (with replacement), so that the number of distinct incident tests concentrates around L with high probability.

Row Weights.

For a fixed test, the number of participating items follows

$$\mathrm{Binomial}\left(n, \frac{L}{m}\right),$$

so that the expected number of defectives in a test equals

$$k\frac{L}{m} = \nu.$$

Poisson Approximation.

In the sparse regime, the number of defectives per test is well approximated by a Poisson random variable with mean ν. Standard concentration bounds apply and are used in the proofs of the main threshold results.

B Equivalence of the Counting Bound and Fano-Based Converse

In this appendix, we show that the classical combinatorial counting bound coincides with the information-theoretic converse derived via Fano's inequality in the noiseless non-adaptive setting.

Setup.

Let S denote the defective set, assumed uniformly distributed over all $\binom{n}{k}$ subsets of $[n]$. Let $Y \in \{0,1\}^m$ be the outcome vector and $\hat{S}$ an estimator of S. Let P_e denote the average error probability.

Fano's Inequality.

Fano's inequality states that

$$H(S|Y) \leq 1 + P_e \log_2 \binom{n}{k}.$$

If $P_e \to 0$, then

$$\frac{H(S|Y)}{\log_2 \binom{n}{k}} \to 0.$$

Lower Bound on Mutual Information.

Using

$$I(S;Y) = H(S) - H(S|Y),$$

and noting that

$$H(S) = \log_2 \binom{n}{k},$$

we obtain

$$I(S;Y) \geq (1 - o(1)) \log_2 \binom{n}{k}.$$

Upper Bound on Mutual Information.

Since Y is an m-dimensional binary vector,

$$H(Y) \leq m,$$

and therefore

$$I(S;Y) \leq H(Y) \leq m.$$

Conclusion.

Combining the lower and upper bounds yields

$$m \geq (1 - o(1)) \log_2 \binom{n}{k}.$$

Hence, the information-theoretic converse derived via Fano's inequality coincides asymptotically with the classical combinatorial counting bound.

Use of AI Tools

The author used ChatGPT (OpenAI) as a writing assistant during the preparation of this seminar paper. The tool was used to improve clarity of exposition, refine structure, and check the consistency of notation and formatting. All mathematical statements, proofs, and technical claims were independently verified by the author.

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